DILPREET KAUR

MAL 1020

Linear Algebra & Differential equations

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Linear Algebra

Motivation

System of Linear equations

$$\left.\begin{aligned} x - y &= 1 \\ x + y &= 3 \end{aligned}\right\} \rightarrow \text{Adding}$$

$2x = 4$

$x = \frac{4}{2} = 2$

$2 + y = 3$

$y = 3 - 2 = 1$

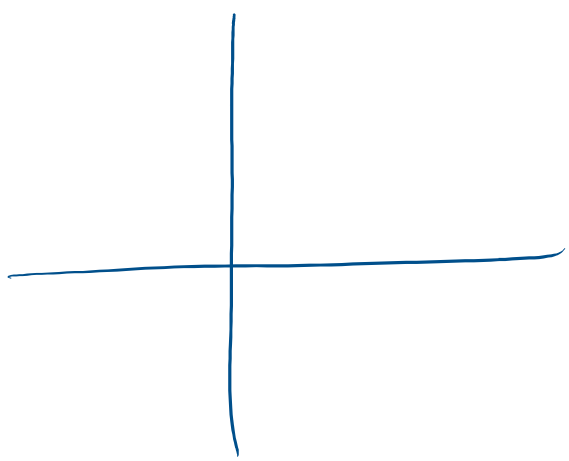
$\boxed{x = 2}$ $\boxed{y = 1}$

Elimination and substitution method.

Graphical Method

$x - y = 1$

$x + y = 3$



(2)

$x - y + z = 2$ — (1)

$\ldots = 4$ — (2)

$$x + y + z = 4 \quad \text{— ②}$$

$$x + y - z = 2 \quad \text{— ③}$$

① + ③ $\quad 2x = 4 \quad \boxed{x = 2}$

① + ② $\quad 2x + 2z = 6 \Rightarrow 4 + 2z = 6$ $\quad \boxed{z = 1}$

③ $\Rightarrow \quad 2 + y - 1 = 2 \Rightarrow \boxed{y = 1}$

$$n \longrightarrow \boxed{\frac{1}{3} n^3}$$

Matrices

Matrix

$$\begin{matrix} a_{11} & a_{12} & \text{—} & a_{1m} \\ a_{21} & a_{22} & a_{23} \text{—} & a_{2m} \\ | & | & & | \\ a_{n1} & \text{—}\; a_{n2} & \text{—} & a_{nm} \end{matrix}$$

a_{23} → Third Row ↓ 2nd Row

$n \times m$ → No of columns ↓ No of Rows.

Add two matrices

$$A_{n \times m} + B_{n \times m} = C_{n \times m}$$

$$a_{ij} + b_{ij} = c_{ij}$$

~~Matrix~~ Scalar multiplication $\lambda(A_{n\times m}) = \lambda A_{n\times m}$

$a_{ij} \longrightarrow \lambda a_{ij}$ (λA)

Matrix Multiplication

$$A_{n\times m} \times B_{m\times l}$$

$$= C_{n\times l}$$

$$\underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}}_{2\times 3} \underbrace{\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}}_{3\times 2} = \underbrace{\begin{pmatrix} 22 & 28 \\ 49 & 64 \end{pmatrix}}_{2\times 2}$$

$\| A$, $\to B$

$1\times1+2\times3+3\times5$

1 row × 1 column

1 row × 2 column

2 row, 1 column

2 row, 2 column

$$\underbrace{AB}_{2\times 2} \neq BA$$

Matrix multiplication is not commutative

$$A+B = B+A$$

Associative

$$A(BC) = (AB)C$$

$A_{n\times m}(BC)_{m\times k}$

$B_{m\times l}$ $C_{l\times k}$

$$\left(A_{n\times m}\ (B_{m\times l}\ C_{l\times k})\right) \qquad (\ )_{n\times m} \to A(BC)_{n\times k}$$

$$(AB)_{n\times l}\, C_{l\times k} = [(AB)C]_{n\times k}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$$

$(AB)_{12}$

$(AB)_{11}$

$$AB = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{21} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}a_{32} \end{pmatrix}$$

22

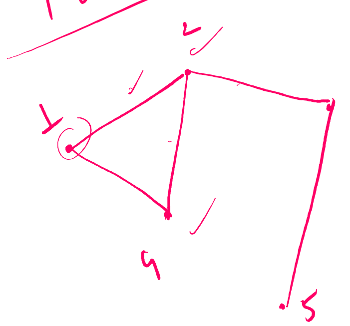
21

$$(AB)_{ij} = \sum_{k=1}^{3} a_{ik} b_{kj}$$

Matrix multiplication is Associative

3×3

Face book



$$\begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 & 1 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 & 0 \\ 5 & 0 & 0 & 1 & 0 & 0 \end{array} = A$$

$$A^2 = \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \end{array}$$
$$\begin{bmatrix} 2 & 1 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Linear Equations

Linear Equations

$$x - y = 1$$
$$x + y = 3$$

$AX = B$

$X = A^{-1}B$

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$AX = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - y \\ x + y \end{bmatrix}$$

$\boxed{AX = B}$

System of Linear Equations

① Graphical method
② Elimination & substitution method

→ Not efficient if $n >>$

$AX = B$
- B → column of constants
- X → Column of variables
- A → matrix of coefficients

$$a_{11}x_1 + a_{12}x_2 + \text{———} + a_{1m}x_m = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \text{———} + a_{2m}x_m = b_2$$
$$\vdots \qquad\qquad \vdots \qquad\qquad \vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \text{———} + a_{nm}x_m = b_n$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \text{——} & a_{1m} \\ a_{21} & a_{22} & \text{——} & a_{2m} \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \end{bmatrix}$$

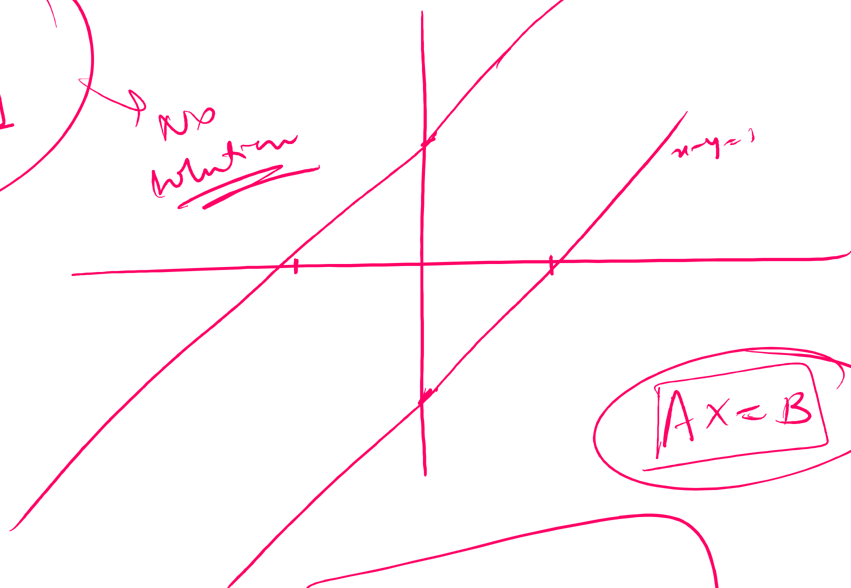
$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \quad X = \begin{bmatrix} x_2 \\ \vdots \\ x_m \end{bmatrix} \quad B = \begin{bmatrix} b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$\boxed{AX = B} \longrightarrow$ How to solve it.

$$x - y = 1$$
$$y - x = 1$$

$\longrightarrow$ No solution

$0 = 2$ ~~X~~



$\boxed{AX = B}$

$$x - y = 1 \qquad \longrightarrow \qquad x = y + 1$$
$$2x - 2y = 2$$

$(1,2)$, $(2,3)$, $(3,4)$ — — —

Rank of a matrix

Row Echelon form

$$\begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{bmatrix} \boxed{0} & \cancel{1} & \cancel{2} & \cancel{3} & \cancel{5} & \cancel{6} \\ 0 & 0 & * & & & \\ 0 & 0 & 0 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

① All rows consisting of only zeroes are at the bottom.

② The leading coefficient (first non-zero entry) of any non-zero row is to right of the leading coefficient of the row above it.

... non-zero Rows

is to right of ...
the row above it.

Rank of matrix $\rightarrow$ Number of Non-zero Rows in Row Echelon form of the matrix.

Elementary Row transformations.

$$R_i \longleftrightarrow R_j$$
$$R_i \longrightarrow \alpha R_i, \quad \alpha \neq 0$$
$$R_i \longrightarrow R_i + \lambda R_j, \quad \lambda \neq 0$$

Find Rank
$$\begin{bmatrix} 5 & 3 & 8 \\ 0 & -1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$R_1 \leftrightarrow R_3$
$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 5 & 3 & 8 \end{bmatrix}$$

$R_3 \rightarrow R_3 - 5R_1$
$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 8 & 8 \end{bmatrix}$$

$R_3 \rightarrow R_3 + 8R_2$
$$\begin{bmatrix} \boxed{1} & -1 & 0 \\ 0 & \boxed{-1} & -1 \\ 0 & 0 & \boxed{0} \end{bmatrix}$$

$\boxed{\text{Rank} = 3}$

Rank = 2

coefficient $\quad AX = B \quad$ variable $\quad$ constants

$(m,1)$ $\quad$ $(n,1)$ $\quad$ (n,m)

coefficient

$$\underbrace{A}_{(n,m)}\ \underbrace{X}_{(m,1)} = \underbrace{B}_{(n,1)}$$

Augmented Matrix $\longrightarrow$ $[A : B] \rightarrow (n, m+1)$

$x + y + z = 4$
$x - y + z = 2$
$x + y - z = 2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & -1 & 2 \end{array}\right]$$

$AX = B$

1. Calculate Rank$(A:B)$
2. Calculate Rank(A)
3. If Rank$(A:B) \neq$ Rank$(A) \Rightarrow$ No solution.
4. Rank$(A:B) =$ Rank$(A) \Rightarrow$ Solution exists

Rank$(A) =$ No. of variables $\rightarrow$ Unique solution

Rank$(A) \neq$ No. of variables $\rightarrow$ Infinitely many solutions.

$$\begin{bmatrix} 1 & -1 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & -1 & 2 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & -2 & 0 \end{bmatrix}$$

$\downarrow R_3 \to R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & -2 & -2 \end{array}\right]$$

Rank$(A:B) = 3$
Rank$(A) = 3$
No. of variables $= 3$
Unique solution

$\ldots y + z = 2$

$x - y + z = 2$

$2y = 2 \rightarrow y = 1$

$-2z = -2 \rightarrow z = 1$

$x = 2$

II

$$\boxed{\begin{aligned} 5x + 3y + 8z &= 16 \\ y + z &= 2 \\ x - y &= 0 \end{aligned}}$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 8 & 16 \\ 0 & 1 & 1 & 2 \\ 1 & -1 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 5 & 3 & 8 & 16 \end{array}\right] \xrightarrow{R_3 \to R_3 - 5R_1} \left[\begin{array}{cccc} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 8 & 8 & 16 \end{array}\right]$$

$R_3 \to R_3 - 8R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\left.\begin{aligned} \text{Rank}[A:B] &= 2 \\ &\shortparallel \\ \text{Rank}(A) &= 2 \end{aligned}\right\} \rightarrow$ Solution exists.

$\text{Rank}(A) = 2 \neq$ No. of variables $= 3$

Infinitely many solution

$x = y \leftarrow x - y = 0$

$2 - y = z \leftarrow y + z = 2$

$\boxed{(y, y, 2-y)}$

y

$(0, 0, 2), (1, 1, 1), (2, 2, 0)$ — — —

Vector spaces $\mathbb{R}^n$

$\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$

$$\mathbb{R}^3 = \{ (x, y, z) \mid x, y, z \in \mathbb{R} \}$$

Addition $\swarrow$ $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2,\ y_1 + y_2,\ z_1 + z_2)$

Scalar multiplication $\swarrow$ $\lambda (x_1, y_1, z_1) = (\lambda x_1, \lambda y_1 + \lambda z_1)$

① $v_1 = (x_1, y_1, z_1) \qquad v_2 = (x_2, y_2, z_2)$

$$\lambda (v_1 + v_2) = \lambda v_1 + \lambda v_2 \checkmark$$

② $\checkmark$ $(\lambda + \beta) v = \lambda v + \beta v$

③ $\checkmark$ $(\lambda \beta) v = \lambda (\beta v)$

④ $\checkmark$ $1 v = v, \qquad 0 \cdot v = \underset{(0,0,0)}{\vec{0}}$

$M_2(\mathbb{R}) \longrightarrow$ set of 2×2 matrices with real entries.

Matrix Addition $\swarrow$
$$\begin{bmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & \delta_1 \end{bmatrix} + \begin{bmatrix} \alpha_2 & \beta_2 \\ \gamma_2 & \delta_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 & \beta_1 + \beta_2 \\ \gamma_1 + \gamma_2 & \delta_1 + \delta_2 \end{bmatrix}$$

Scalar mul $\swarrow$
$$\lambda \begin{bmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & \delta_1 \end{bmatrix} = \begin{bmatrix} \lambda \alpha_1 & \lambda \beta_1 \\ \lambda \gamma_1 & + \lambda \delta_1 \end{bmatrix}$$

$$P_n(x) = \{ a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_i \in \mathbb{R} \}$$

$f(x) + g(x) =$

$\lambda f(x)$

Evaluation Scheme

Mid semester Ex $\longrightarrow$ 25%

Major Ex —— 45%

quizzes —— 20% (Best two)

Assignments / Tutorial $\longrightarrow$ 10%